Lachlan Watts-Tobin

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EDUCATION

University of California, Berkeley Berkeley, CA
B.S. Electrical Engineering & Computer Science (GPA: 3.3 / 4.0) May 2028
Coursework: Digital Design and Integrated Circuits (with ASIC Lab), Computer Architecture, Internet of Things, Machine Structures, Circuits and Devices, Data Structures, Signals and Information Processing, Solid Mechanics

SKILLS

Languages: SystemVerilog, Verilog, C, C++, CUDA, Python, Chisel, Scala, RISC-V, Tcl, Perl, MATLAB
Platforms: Cadence (Innovus, Indago), Synopsys (VCS, Verdi), UVM, Linux, Git, PyTorch, TensorRT, Triton

EXPERIENCE

Pano AI San Francisco, CA
Hardware Engineering Intern May 2026 – August 2026
- Redesigned a power-management system for an experimental unit improving battery life and remote cycling.
- Designed an edge-computer system layout optimizing for space efficiency and thermal distribution.
- Authored standard operating procedures and designed a manufacturing plan to scale edge-computer production.

Vision IV (Remote) Yokohama, Japan
Hardware Engineering Intern May 2025 – July 2025
- Optimized a high-performance electrical component using experimental materials to improve reliability.
- Analyzed potential failure points to guide data-driven improvements to component performance.

Theoretical and Applied Fluid Dynamics Laboratory Berkeley, CA
Undergraduate Researcher Sept 2026 – Present
- Design, bring up, and optimize microcontrollers for a swarm of autonomous sailboats (USVs) for ocean sensing.
- Conduct on-site field testing and live debugging of boat electronics during on-water deployments.

PROJECTS

CS61CPU
UC Berkeley, College of Engineering, (November 2025 – December 2025)
- Designed a 5-stage pipelined CPU in Logisim capable of executing 32-bit RISC-V instructions.

Watts & Cseh Electronics
Co-Founder & Lead Electrical Engineer (May 2026 – Present)
- Created and verified analog circuits for guitar effects pedals, inspired by vintage hardwired models.

AI-Powered Guitar Pedal
Hackathons at Berkeley, (June 20, 2026)
- Bridged AI to hardware, pushing generated parameters from plain English to an ESP32 DSP device over WiFi.

Automated Trading Card Sorter
UC Berkeley, College of Engineering, (April 2026 – May 2026)
- Engineered a machine that automatically sorts trading cards by their live market value, using CV and scraping.

Self-Hosted Portfolio Website
Personal, (May 2026 – August 2026)
- Self-hosted a portfolio site on a Raspberry Pi via a tunnel with custom DNS and automated Git deploys.

Filament Winder G-code Slicer
UC Berkeley, College of Engineering, (January 2025 – April 2025)
- Developed standalone software to generate G-code from part files, visualize it, then send it to a custom winder.

EXTRACURRICULARS

Bike Builders of Berkeley Berkeley, CA
Lead Electrical Engineer August 2024 – Present
- Lead the electrical design of the e-bike's EV drivetrain, including custom PCBs and power electronics.
- Balance workloads across website, firmware, and hardware as project manager.

UC Berkeley, College of Engineering Berkeley, CA
Undergraduate Tutor July 2025 – Present
- Tutor lower-division electrical engineering and computer science coursework to students every weekday.